

Kushagra Dixit

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Software Engineer with 2 years of experience building scalable, fault-tolerant backend systems using microservice architecture. Skilled in Python, C++, and Java with expertise in distributed systems and database-driven applications. Focused on delivering stable, secure, and high-performance platforms.

Experience

Software Development Engineer - OYO

Aug 2021 – Sep 2022

- Developed **RESTful APIs** using **Java**, **Spring Boot**, and **PostgreSQL**, capable of handling **90,000+** requests per minute, ensuring high availability and scalability with **Kubernetes** and **Docker**.
- Introduced parallelized microservice calls using **Java multithreading** to slash **P99 latency of pricing aggregator by 1s**, leveraging **MongoDB** and **Kafka** for distributed caching and synchronization, **cutting server load by 40%**.
- Enhanced hotel pricing retrieval via **SQL** indexing and caching, **cutting average query time by 35% (50 ms)** and **increasing booking conversion by 10%** across distributed microservices.

Research and Development Engineer - Synopsys

Sep 2022 – Aug 2023

- Developed core **C++** modules for **DFT (Design for Test)** automation, transforming **RTL logic into gate-level netlists** to enable **scalable chip synthesis**.
- Implemented **Design Rule Checking** tools at the **RTL level** specifically for **inserted DFT logic**, enabling early detection of integration faults and reducing **design iteration cycles by 20%**.
- Debugged and resolved critical DFT insertion and synthesis issues** in customer designs using **GDB** and Linux-based tooling, and **automated validation workflows** with shell scripting, improving product stability.

ML Research Programmer (Internship) – University of Pennsylvania

May 2024 – Aug 2024

- Designed **SEAR**, an **adaptive prompting framework** for tabular reasoning with LLMs. Built scalable inference and evaluation pipelines using **Python**, **LangChain**, **Docker**, and **scikit-learn**, improving **retrieval-augmented generation** performance by **5%** over baselines.
- Converted semi-structured tables (e.g., Wikipedia infoboxes) into structured representations for temporal QA by creating **knowledge graphs** in **Neo4j** and generating canonical keys to build normalized tables.
- Fine-tuned LLMs using **LoRA** with **PyTorch** and **Hugging Face Transformers** to improve **SQL generation**, achieving **12–15% gains** in accuracy across models.

Projects

TactiXpert | (*Winner - VCT Hackathon organized by AWS & Riot Games*)

[LINK]

- Created a **multi-agent LLM-based chatbot** to help Valorant coaches form pro esports teams, with agents handling dialogue, strategy logic, and **LLM-driven player retrieval** from **DynamoDB**.
- Designed the agent system using **AWS Bedrock** and developed a modular backend with **Python**, **FastAPI**, **LangChain**, and **Docker**, integrated with a **React.js** frontend.

Smart Parking and Anti-theft System | ETI Labs

[LINK]

- Engineered a **multi-threaded Python backend** on Raspberry Pi for a **smart parking system**, integrating real-time user identification via **OpenCV** and a **PyQT** based UI for local control.
- Developed a web-based frontend with **JavaScript**, **HTML**, **CSS**, and **PHP**, improving parking efficiency and user experience.

TrajectIQ | University of Utah

[LINK]

- Orchestrated a **BERT-based trajectory imputation system** with end-to-end design. Trained the model on trajectory data using a custom pipeline built with **Apache Spark**.
- Delivered scalable data processing and a user-friendly interface, achieving an **F1-Score of 0.81** for imputed coordinates.

Technical Skills

Languages: C++, Python, Java, JavaScript, HTML, CSS, PHP, SQL, Bash

Frameworks and Libraries: Spring-Boot, Micro-service Architecture, RESTful APIs, Apache Spark, Docker, Apache Kafka, Kubernetes, FastAPI, GDB, React, Git, Perforce

Cloud and Databases: AWS, Unix/Linux, PostgreSQL, MongoDB, DynamoDB, Redis, MySQL

Education

University of Utah

Aug 2023 – May 2025

M.S. Computer Science

GPA: 3.97/4.0

Netaji Subhas Institute of Technology

Aug 2017 – May 2021

B.E. Instrumentation and Control

GPA: 8.26/10

Publications

- Enhancing Temporal Understanding in LLMs for Semi-structured Tables (Published – **NAACL 2025**) [LINK]
- LLM-Symbolic Integration for Robust Temporal Tabular Reasoning (Accepted – **ACL 2025**) [LINK]
- Adaptive Prompting: A Dynamic Approach to Temporal Table Reasoning (Under Review – **EMNLP 2025**) [LINK]